Tech Stack

* OOPs → object oriented Programming :-

Procedure oriented	object oriented..
Procedural programming is a programming paradigm where program is organized as a sequence of function or procedure that operates on data	* object oriented programming (OOP) is a programming paradigm where the program is organized around objects that contain both data (attributes) and function (methods).
* Procedural programming focuses on function	* object oriented programming focuses on object

* Before OOP existed, most program were written using procedural programming language like 'c'...

In procedural programming :-

- The program is divided into functions
- Data and function are seperate
- function operate on the data..

As software became large & complex, procedural programming started causing problems:-

- code become difficult to manage
- Data security weak.
- large programs become hard to maintain..

To solve all this issues, object oriented programming was introduced..

OOP organizes program using objects that combine data and behaviour together.. This makes programs easier to learn, design & maintain..

Real world Analogy:

Think about a CAR

Data (Attributes)

- Color
- Speed
- fuel level

Behavior (methods)

- start()
- Accelerate()
- Break()

An oops made program

- data (color, speed)
- functions (start, stop).

Procedure programming would separate these things into different functions & variables.

Program
↓
functions
↓
Data..

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Example in C:

{procedural}.

```
#include <stdio.h>

int balance = 1000;

void deposite (int amount)
{
    balance = balance + amount
}

void withdraw (int amount)
{
    balance = balance - amount;
}

int main()
{
    deposit(500); # 1000 + 500 ⇒ 1500
    withdraw(200); # 1500 - 200 ⇒ 1800
    print ("Balance: %d", balance); # 1800
```

Explination

Here:-

- balance is global data
- deposit() & withdrawal() are function
- function directly modify the data..

Problem

any function can modify balance.

This can cause security & maintenance problems..

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Object-oriented programming

Program
↓
class
↓
objects
↓
Methods.

* Data and functions are combined together in idea class
* Access to data is controlled..

Example python

```
class BankAccount:
    def __init__(self, balance):
        self.balance = balance

    def deposit(self, amount):
        self.balance = self.balance + amount

    def withdraw(self, amount):
        self.balance = self.balance - amount


account = BankAccount(1000)

account.deposit(500)
account.deposit(200)
print("Balance:", account.balance) //
```

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Here

- Bank account class
- balance is data
- deposit() & withdraw() are methods
- Every is organized inside the class

This make:-

- Organized
- secure
- easy to maintain

Procedural programming	object oriented programming.
focus is on functions	focus is an object
Data & functions are seperate	Data & functions are combined
Less secure	More secure
Difficult to maintain large program	Easier to manage large programs
code reuse is limited	code reuse is possible
C, Pascal	Java, Python, C++, C#
Better modularity,	security & code reuse

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Common mistake students make:

Many students think OOP mean only classes & objects

That is wrong.

OOP actually mean:

- Encapsulation
- Inheritance
- polymorphism
- Abstraction..

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